



Blockchain dApp Development (UQ24CA343AA3)

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Unit II- Consensus Mechanisms and Smart Contracts

Consensus Mechanisms: Proof of Elapsed Time, Proof of Authority, Proof of Importance

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- Consensus mechanism where each participating node waits for a randomly assigned waiting time.
- The node with the shortest waiting time is selected to create the next block. **Example: Hyperledger Sawtooth**

Key Points

- Random waiting time determines the winner.
- No mining or staking required.
- Energy efficient.
- Suitable for permissioned blockchain networks.

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Proof of Elapsed Time (PoET)



How It Works

1. Each node receives a random waiting time.
2. Nodes wait for their assigned time.
3. The node whose timer expires first creates the new block.
4. Other nodes verify the block.
5. The block is added to the blockchain.

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Proof of Elapsed Time (PoET)



Advantages

- Low power consumption.
- Fair and random block selection.
- Fast block creation.

Limitations

- Requires trusted hardware (e.g., Intel SGX).
- Mainly suitable for private/enterprise blockchains.

Consensus mechanism in which pre-approved and trusted validators are responsible for verifying transactions and creating new blocks.

Key Points

- Identity-based validation.
- No mining or staking.
- High transaction speed.
- Used in private and consortium blockchains.

How It Works

1. Trusted validators are selected.
2. Validators verify transactions.
3. One validator creates the block.
4. The block is shared with the network.
5. Other validators confirm the block.

Advantages

- Fast transaction processing.
- Low computational cost.
- High scalability.
- Suitable for enterprise applications.

Limitations

- Less decentralized.
- Requires trust in validators.
- Validator selection is controlled.

Consensus mechanism that selects validators based on their overall contribution to the blockchain network rather than only the amount of cryptocurrency they own.

Selection Factors

- Amount of cryptocurrency held.
- Transaction activity.
- Network participation.
- Reputation and contribution.

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Proof of Importance (Pol)

Key Points

- Rewards active participants.
- Encourages regular transactions.
- Prevents wealth-based domination.
- Improves network participation.

How It Works

1. Users hold cryptocurrency.
2. The network calculates an importance score.
3. Users with higher scores are more likely to validate blocks.
4. Validators receive rewards.

Advantages

- Promotes active network usage.
- Fairer than pure staking.
- Encourages decentralization.

Limitations

- More complex calculation.
- Less widely adopted.
- Requires continuous network activity.

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Proof of Importance (Pol)

Feature	PoET	PoA	Pol
Selection Method	Random Waiting Time	Trusted Validators	Importance Score
Energy Consumption	Very Low	Very Low	Low
Mining Required	No	No	No
Decentralization	Medium	Low	High
Best Suited For	Enterprise Blockchain	Private/Consortium Blockchain	Public Blockchain
Example	Hyperledger Sawtooth	VeChain	NEM

Key Takeaways

- Proof of Elapsed Time (PoET) selects validators based on a randomly assigned waiting time.
- Proof of Authority (PoA) relies on trusted validators to secure the network.
- Proof of Importance (PoI) rewards participants based on their contribution and activity.
- Each consensus mechanism is designed to balance security, efficiency, scalability, and decentralization according to different blockchain requirements.

Unit II- Consensus Mechanisms and Smart Contracts

Consensus Mechanisms: Proof of Work, Proof of Stake, Proof of Activity, Proof of Burn

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Protocol that enables all nodes in a blockchain network to agree on the validity of transactions and maintain a single, consistent version of the ledger without a central authority.

Why is Consensus Needed?

- Validates blockchain transactions.
- Prevents double spending.
- Ensures all nodes maintain the same ledger.
- Builds trust in decentralized networks.
- Protects the blockchain from malicious activities.

Steps

1. A transaction is created.
2. The transaction is broadcast to the network.
3. Nodes verify the transaction.
4. Consensus mechanism validates the transaction.
5. A new block is created.
6. The block is added to the blockchain.
7. The ledger is updated across all nodes.

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Types of Consensus Mechanisms

- Proof of Work (PoW)
- Proof of Stake (PoS)
- Proof of Activity (PoA)
- Proof of Burn (PoB)
- Proof of Elapsed Time
- Proof of Authority
- Proof of Importance

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Proof of Work (PoW)



- Proof of Work is a consensus mechanism in which miners compete to solve complex mathematical puzzles.
- The first miner to solve the puzzle earns the right to add the next block and receives a reward.

Key Points

- Highly secure.
- Requires significant computational power.
- High energy consumption.
- Used in Bitcoin.

Working

1. Transactions are collected.
2. Miners solve a cryptographic puzzle.
3. First successful miner creates the block.
4. Other nodes verify the block.
5. The block is added to the blockchain.

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Proof of Work (PoW)

Advantages

- Strong security.
- Resistant to attacks.
- Proven and reliable.

Limitations

- High electricity consumption.
- Lower transaction speed.
- Expensive mining hardware.

Proof of Stake selects validators based on the amount of cryptocurrency they stake rather than computational power.

Key Points

- Energy efficient.
- Faster than PoW.
- No mining required.
- Used in Ethereum 2.0.

Advantages

- Low energy usage.
- Faster block creation.
- Lower hardware requirements.

Limitations

- Wealth concentration concerns.
- Requires sufficient stake.
- Validator selection depends on network rules.

- Combines Proof of Work and Proof of Stake.
- Mining begins with PoW, and block validation is completed by selected stakeholders.

Key Points

- Hybrid consensus mechanism.
- Improves security.
- Reduces reliance on mining alone.
- Balances energy usage and decentralization.

Working

1. Miners generate an empty block using PoW.
2. Validators are selected based on stake.
3. Validators verify the block.
4. Block is finalized and added.

Advantages

- Better security.
- Shared responsibility.
- Reduced attack risk.

Limitations

- More complex implementation.
- Requires both miners and validators.

Proof of Burn allows participants to permanently destroy (burn) a portion of their cryptocurrency to gain the right to validate transactions and create new blocks.

Key Points

- No expensive mining hardware.
- Demonstrates commitment to the network.
- Energy efficient.
- Supports long-term participation.

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Proof of Burn (PoB)



Working

1. Coins are sent to an unspendable address.
2. Burned coins prove long-term commitment.
3. Validator gains mining rights.
4. Validator creates new blocks and earns rewards.

Advantages

- Low energy consumption.
- Encourages network commitment.
- Reduced hardware requirements.

Limitations

- Permanent loss of coins.
- Less commonly adopted.
- Risk associated with burning assets.

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Comparison of Consensus Mechanisms

Feature	PoW	PoS	Proof of Activity	PoB
Validation	Mining	Staking	Mining + Staking	Burning Coins
Energy Usage	High	Low	Medium	Low
Hardware Requirement	High	Low	Medium	Low
Security	High	High	Very High	Moderate
Example	Bitcoin	Ethereum	Decred	Slimcoin

- Consensus mechanisms enable trust in decentralized blockchain networks.
- Different mechanisms balance security, energy efficiency, and performance.
- Proof of Work relies on mining.
- Proof of Stake relies on staking.
- Proof of Activity combines mining and staking.
- Proof of Burn relies on permanently burning cryptocurrency.

Unit II- Consensus Mechanisms and Smart Contracts

Introduction to smart contracts, Smart contracts platforms, structure

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Smart Contracts



- Self-executing digital agreement stored on a blockchain.
- Terms and conditions are automatically executed when predefined conditions are met.

Key Points

- Digital agreement stored on the blockchain.
- Automatically executes predefined conditions.
- No intermediary required.
- Transparent and tamper-resistant.
- Uses blockchain technology for security.

Real-World Example

- Online ticket booking
- Insurance claim processing
- Automatic payment after product delivery

Problems with Traditional Contracts

- Depend on intermediaries.
- Time-consuming.
- Higher transaction costs.
- Possibility of fraud.
- Manual execution.

Smart Contracts Solve These Problems By

- Automating execution.
- Reducing costs.
- Increasing transparency.
- Improving security.
- Eliminating intermediaries.

Working Steps

1. Contract terms are written as code.
2. The contract is deployed on the blockchain.
3. Users interact with the contract.
4. The blockchain verifies the conditions.
5. The contract executes automatically.
6. Results are permanently recorded on the blockchain.

Once deployed, smart contracts execute automatically without human intervention.

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Features of Smart Contracts

- Automated execution
- Immutable
- Transparent
- Secure
- Decentralized
- Cost-effective
- Trustworthy

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Smart Contracts Platforms



Smart contract platforms provide the blockchain infrastructure where smart contracts are deployed, executed, and stored securely.

Popular Platforms

Ethereum – Most widely used platform for smart contract development.

BNB Smart Chain – Supports fast and low-cost smart contract execution.

Hyperledger Fabric – Enterprise blockchain platform for business applications.

Popular Platforms

Solana – High-performance blockchain with fast transaction processing.

Cardano – Blockchain platform focused on security and scalability.

Polygon – Layer-2 platform that improves Ethereum scalability and reduces transaction costs.

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Smart Contracts Development Tools



Development tools help developers write, compile, test, deploy, and interact with smart contracts before deploying them to a blockchain platform.

Development Tools

Remix IDE

Ganache

MetaMask

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Smart Contracts Development Tools-Remix IDE

- Web-based Integrated Development Environment (IDE)
- Written in Solidity
- Write, compile, debug, deploy, and test Ethereum smart contracts.

Key Features

- Browser-based (no installation required).
- Supports Solidity programming.
- Built-in Solidity compiler.
- Deploy and test smart contracts.
- Debugging and error detection.
- Connects with MetaMask and Ganache.



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Smart Contracts Development Tools-Remix IDE

Uses

- Writing Solidity code.
- Compiling smart contracts.
- Testing contracts.
- Deploying contracts.

Advantages

- Easy for beginners.
- Free to use.
- No setup required.
- Supports plugins and debugging.

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Smart Contracts Development Tools-Ganache

Personal local blockchain that allows developers to deploy, test, and debug Ethereum smart contracts without using the public Ethereum network.

Key Features

- Creates a private blockchain.
- Generates multiple test accounts.
- Provides free test Ether.
- Simulates Ethereum transactions.
- Fast deployment and testing.



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Smart Contracts Development Tools-Ganache



Uses

- Local blockchain development.
- Testing smart contracts.
- Debugging transactions.
- Experimenting without real money.

Advantages

- No transaction fees.
- Instant block creation.
- Safe testing environment.
- Easy to reset blockchain state.

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Smart Contracts Development Tools-MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to manage Ethereum accounts and interact with blockchain applications (DApps).

Key Features

- Stores Ethereum accounts securely.
- Manages public and private keys.
- Connects to blockchain networks.
- Signs blockchain transactions.
- Interacts with DApps.



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Smart Contracts Development Tools-MetaMask

Uses

- Create blockchain wallet.
- Send and receive Ether.
- Connect to Remix IDE.
- Deploy smart contracts.
- Access decentralized applications.

Advantages

- Easy to install.
- User-friendly interface.
- Supports multiple Ethereum networks.
- Secure wallet management.

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Comparison Table

Feature	Remix IDE	Ganache	MetaMask
Type	Web IDE	Local Blockchain	Crypto Wallet
Purpose	Write & Compile Contracts	Test Blockchain	Manage Accounts & Transactions
Installation	No (Web-based)	Yes	Browser Extension / Mobile App
Main Users	Developers	Developers	Developers & End Users
Cost	Free	Free	Free

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Real-World Development Workflow



1. **Write** the smart contract in **Remix IDE**.
2. **Compile** the Solidity code in Remix.
3. **Connect MetaMask** to manage your Ethereum account.
4. **Connect MetaMask to Ganache** (local blockchain).
5. **Deploy** the contract from Remix using the MetaMask account.
6. **Interact** with the deployed contract through Remix or a decentralized application (DApp).

Key Takeaways

- Smart contracts automate agreements without requiring intermediaries.
- Blockchain platforms provide a secure environment for executing smart contracts.
- Development tools simplify writing, testing, and deploying smart contracts.
- Smart contracts improve transparency, security, and efficiency in digital transactions.

Unit II- Consensus Mechanisms and Smart Contracts

Development, Deployment & Testing of smart contracts

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Smart Contracts Development



Smart contract development is the process of designing, writing, compiling, and debugging blockchain-based programs that execute automatically when predefined conditions are met.

Key Points

- Smart contracts are written as program code.
- Solidity is the most widely used language for Ethereum.
- Development is performed using Remix IDE or similar IDEs.
- The contract must be error-free before deployment.

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Smart Contracts Development

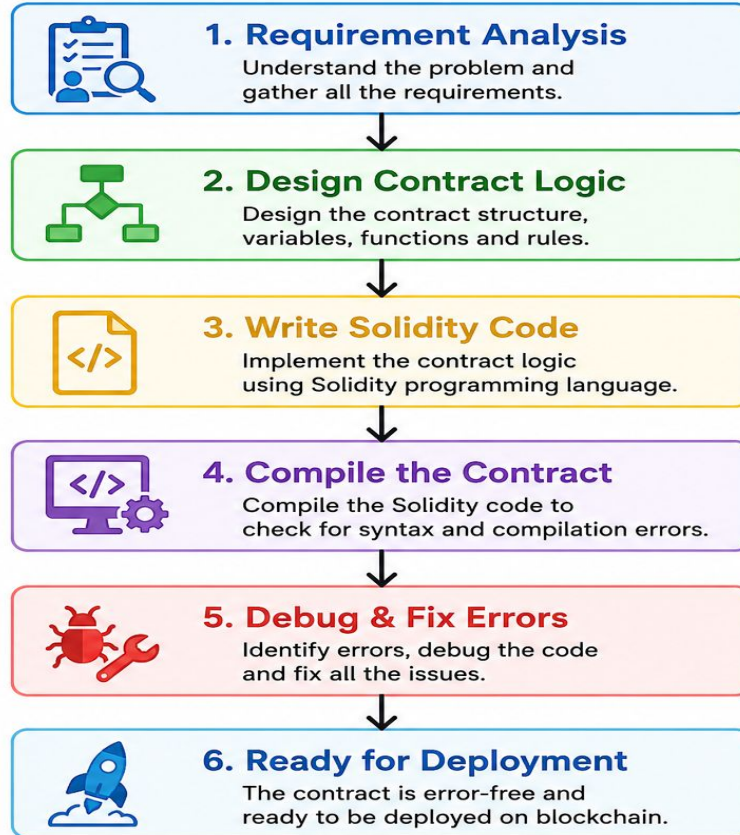


Development Process

- Define business requirements.
- Design the contract logic.
- Write the smart contract using Solidity.
- Compile the contract.
- Fix syntax and compilation errors.

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Smart Contracts Development Workflow



Deployment is the process of publishing a compiled smart contract onto a blockchain network, making it available for execution.

Key Points

- Deployment creates a permanent contract instance.
- Every deployed contract has a unique address.
- Deployment requires blockchain transaction confirmation.

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Deployment of Smart Contracts

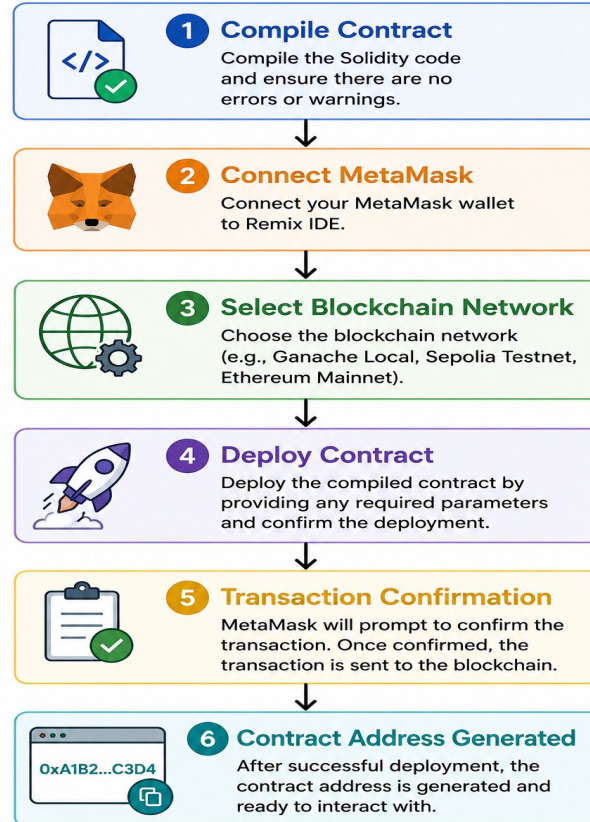


Deployment Process

- Connect MetaMask wallet.
- Select the blockchain network.
- Deploy the compiled contract.
- Confirm the deployment transaction.
- Obtain the contract address.

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Deployment Workflow



Testing is the process of verifying that a smart contract performs correctly, securely, and efficiently before deployment to a production blockchain.

- Detect coding errors.
- Verify contract logic.
- Prevent security vulnerabilities.
- Reduce deployment risks.
- Ensure expected outputs.

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Types of Testing



Functional Testing

- Checks whether functions work correctly.

Unit Testing

- Tests individual functions independently.

Integration Testing

- Tests interaction between multiple contracts.

Security Testing

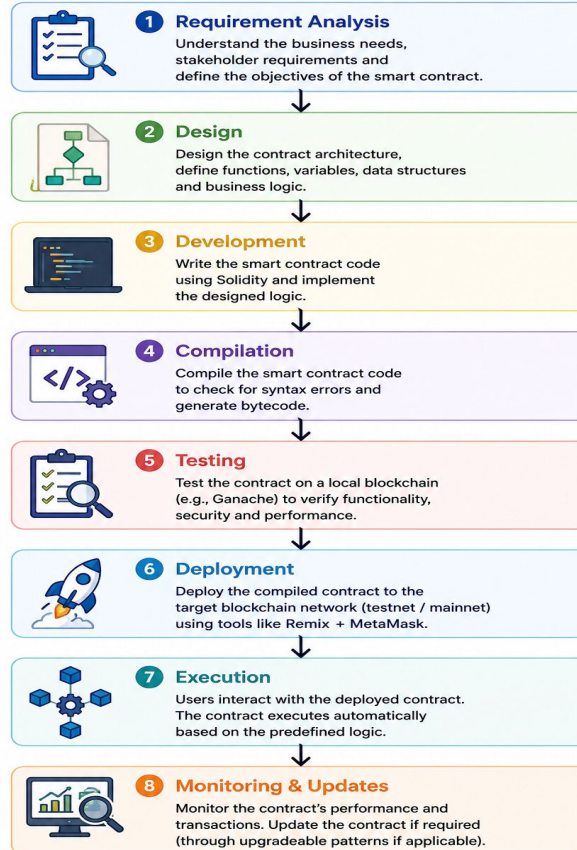
- Identifies vulnerabilities.

Performance Testing

- Evaluates execution efficiency.

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Complete Smart Contract Lifecycle



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Best Practices



During Development

- Write simple and readable code.
- Use meaningful variable names.
- Avoid unnecessary complexity.

During Deployment

- Verify network selection.
- Keep backup of contract information.
- Review deployment parameters.

During Testing

- Test all possible inputs.
- Validate outputs.
- Test security scenarios.

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University Certificate Verification



Development

- Create a smart contract to store certificate details.

Deployment

- Deploy the contract to a blockchain network.

Testing

- Verify certificate upload.
- Check certificate validation.
- Test tamper resistance.

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Key Takeaways

- Development involves designing, coding, and compiling smart contracts.
- Deployment publishes smart contracts on a blockchain network.
- Testing ensures correctness, security, and reliability.
- Remix IDE, Ganache, and MetaMask simplify the development process.

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Use cases Smart contract & case studies

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Use cases of Smart contract

- Banking & Finance
- Supply Chain Management
- Healthcare
- Real Estate
- Insurance
- Education
- Voting Systems
- Digital Identity

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Applications of Smart contract

Industry	Application
Banking	Automated payments and fund transfers
Healthcare	Secure patient record management
Supply Chain	Product tracking and traceability
Real Estate	Digital property registration
Insurance	Automatic claim settlement
Education	Certificate verification
Government	Digital identity and public records
Voting	Secure electronic voting systems

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Case Study 1 – Supply Chain Management



Problem

Tracking products across multiple suppliers is time-consuming and lacks transparency.

Smart Contract Solution

- Records every stage of the product journey.
- Automatically updates ownership.
- Verifies product authenticity.
- Releases payments after successful delivery.

Benefits

- Improved transparency
- Reduced fraud
- Faster payments
- Better traceability

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Case Study 2 – Education Certificate Verification



Problem

Employers spend time verifying certificates and may encounter fake documents.

Smart Contract Solution

- Universities issue certificates on the blockchain.
- Students receive a secure digital certificate.
- Employers verify certificates instantly.
- Records cannot be altered.

Benefits

- Tamper-proof certificates
- Instant verification
- Reduced paperwork
- Increased trust

Case Study 3 – Insurance Claim Processing

Problem

Traditional insurance claims involve manual verification and long processing times.

Smart Contract Solution

- Claim conditions are defined in a smart contract.
- Required documents are verified automatically.
- If conditions are met, the payment is released automatically.

Benefits

- Faster claim processing
- Reduced fraud
- Lower administrative costs
- Improved customer satisfaction

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Key Takeaways

- Smart contracts automate business processes.
- They eliminate intermediaries and reduce costs.
- Blockchain ensures transparency and security.
- Smart contracts are widely used in finance, healthcare, education, insurance, and supply chain management.



THANK YOU

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